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# Research on hybrid of ALOHA and multi-fork tree Anti-collision algorithm for RFID

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## Abstract

The work aims at the problem of multi-tag collision in RFID system, then analysis the binary tree algorithm and ALOHA and points out their shortcomings, we proposed a Hybrid of ALOHA and Tree (HAMT) Algorithm. The algorithm first uses the DFSA algorithm for tag identification, and dynamically selects the multi-fork tree algorithm for tag identification based on the number of unidentified tags secondly, thus ensuring that the tag is 100% identified, and improves recognition efficiency and shortens recognition time. The results of the simulation show that when the number of tags is about 1000, the recognition efficiency of the HAMT algorithm can reach around 0.72. So the HAMT algorithm can solve the multi-tag collision problem.

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**Keywords:** RFID; Multi-fork tree; ALOHA; tags; Anti-collision algorithm

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## 1. Introduction

Radio frequency identification is a kind of non-contact automatic identification technology that uses radio frequency signals through spatial electromagnetic coupling, and is a key supporting technology for the Internet of Things<sup>[1]</sup>. At present, the development of domestic RFID technology is approaching maturity, and the cost of tag production is continuously decreasing. And its application range is continuously expanding, involving logistics management, anti-counterfeiting, animal management, access control, intelligent identification and other fields<sup>[2]</sup>.

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The composition of an RFID system generally has three parts: tags, readers, and a computer background system<sup>[3]</sup>. The three major technical problems of RFID technology: unified standards, multi-tag collision and information security, where multi-tag collision refers to the signal sent by the reader, there are multiple tags responding at the same time within the signal range, the reader cannot simultaneously recognize multiple tags phenomenon. Therefore, it is of great research significance to propose an effective and fast solution to the collision problem of RFID tags<sup>[4]</sup>. Aiming at the problem of tag collision, two kinds of anti-collision algorithms have been proposed: deterministic algorithm based on binary tree and random algorithm based on ALOHA<sup>[5]</sup>. The former algorithms mainly includes: BST algorithm<sup>[6]</sup>, Dynamic Binary Search Tree (DBST) algorithm<sup>[7]</sup>, Collision Tree (CT) algorithm<sup>[8]</sup>, Adaptive AMS algorithm<sup>[9]</sup>, IACT algorithm<sup>[10]</sup>, ISE-BS algorithm<sup>[11]</sup>, Self-Adjusting Hybrid Tree (AHT) algorithm<sup>[12]</sup>, etc. The latter algorithm mainly includes: SA algorithm<sup>[13]</sup>, FSA algorithm<sup>[14]</sup>, DFSA algorithm<sup>[15]</sup>, etc. Both of these two types of algorithms can solve the problem of multi-tag collisions, but the ALOHA algorithm is random. When the number of tags to be recognized is large, the probability of tag collision increases, and some tags may still be recognized. There is a phenomenon of "starvation" of tags, and although the tree-based algorithm can achieve 100% tag reading rate, but the recognition time is longer.

In this paper, in view of the lots of tags, the recognition efficiency of the DFSA algorithm decreases and there may be shortcomings that the tag can't be recognized by 100% and the multi-tree algorithm recognizes the increase in tag time. A hybrid of ALOHA and multi-tree is proposed (HAMT) algorithm. The algorithm first uses the DFSA algorithm for tag recognition, and dynamically selects the multi-tree algorithm for tag recognition according to the number of unrecognized tags secondly, thereby ensuring 100% tag recognition, improving throughput and shortening recognition time.

## 2. The HMAT algorithm description

### 2.1. The manchester coding

The manchester coding describes logic "0" or logic "1" by the change of level. In the HMAT algorithm, its main function is that the reader can accurately detect the collision position between the tag IDs. Suppose there are two tags, Tag1 and Tag2, and their ID codes are: 0010110 and 00110001. The principle of detecting collision bits is shown in Figure 1:

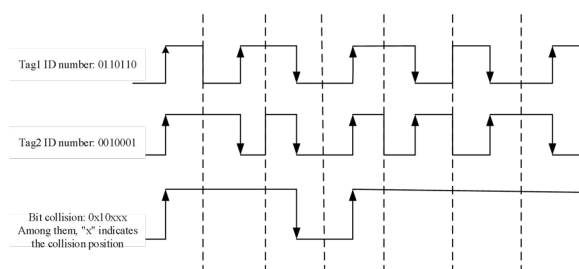


Fig. 1. Manchester coding example

### 2.2. The theory of HAMT algorithm

Aiming at the problems of long recognition time and low throughput of tree based algorithm and Aloha algorithm, the label can not be 100% identified, a Hybrid of ALOHA and Multi-fork Tree (HAMT) algorithm is proposed. The algorithm recognizes tags through two stages: dynamic frame slot stage and multi-fork tree stage.

Dynamic frame time slot stage: according to the tags to be identified, the slot frame length is determined, and select the time slot number that the tag responds to according to the time slot selection rules, and finally determine the state of the time slot, if it collides into the multi-tree stage.

Multi-fork tree stage: Aiming at the collision tags, the multi-fork tree is dynamically selected. The card reader uses Manchester coding principle to detect collision bits and uses the collision bits to obtain the query prefix.

The schematic diagram of its recognition process is shown in Figure 2:

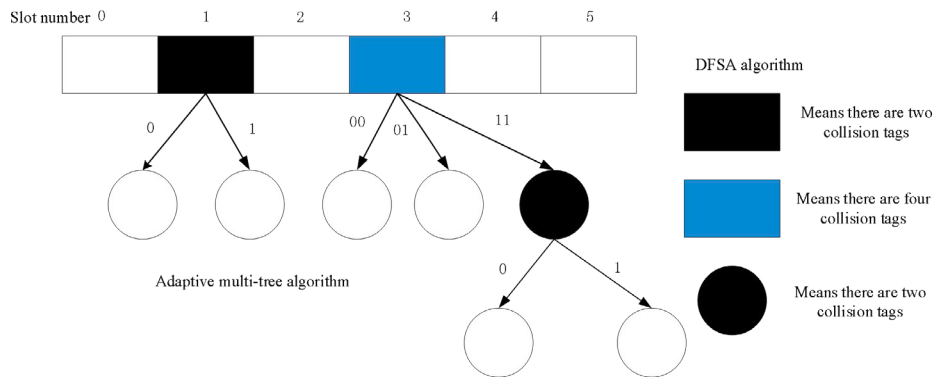


Fig. 2. Hybrid algorithm identification tag process schematic

Figure 2 has a total of 6 tags to be identified. In the first stage, using DFSA algorithm to identify tags, then a total of two collision slots are generated. Among them, there are 2 collision tags in slot 1 and 4 collision tags in slot 3. The collision tags enter the second stage, which uses an adaptive multi-tree algorithm for recognition, that is, two collision tags are identified by binary tree algorithm, and four collision tags are identified by quadtree algorithm. HAMT algorithm can greatly reduce idle time slots, thereby improving system throughput.

### 2.3. Flow of HAMT

For the HAMT,  $i$  is the time slot number, and the Vogt algorithm<sup>[16]</sup> is used to estimate the value of  $N$  of tags to be identified, and DFSA algorithm and adaptive multi-tree algorithm are used for tag recognition until the tags to be recognized is 0, and it ends, the flow chart of HAMT is shown in Figure 3.

The flow of the HAMT are as follows:

Step 1: Using Vogt algorithm to estimate the value of the tags  $N$  to be identified. The literature [16] indicates that the  $L$  and the  $N$  have following relationship:

$$L = 2^{\lceil \log_2 N \rceil} \quad (1)$$

Notes:  $L$  is frame length and  $N$  is total number of tags.

The required frame length can be calculated by formula (1);

Step 2: The card reader sends a Request ( $N$ ,  $L$ ) command, and all tags respond;

Step 3: The tag selects its required  $i$  from  $(0, L-1)$  time slot according to its own ID number;

Step 4: Determine the status of time slot  $i$ , one of the following three cases:

(1) Time slot  $i$  is a successful time slot, that is, there is only one tag response in this time slot, which is a successful identification tag. Go to the next time slot  $i = i + 1$ ;

(2) if  $i$  is an idle time slot, that is, there is no tag response in this time slot, Go to next time slot  $i = i + 1$ ;

(3) if  $i$  is a collision time slot, that is, there are more than two tags responding to this time slot, which are collision tags. These tags go to the next step for processing;

Step 5: Estimate the value of  $n$  of collision tags, and determine whether  $n$  is greater than or equal to 4:

(1) If  $n \geq 4$ , the card reader detects the value of the two highest collision bit, then uses the three-tree algorithm or the four-tree algorithm to identify according to these two values.

(2) If  $n < 4$ , the card reader only detects the value of the highest collision bit, then uses the BT algorithm to identify;

Step 6: Determine whether the tags are all successfully identified;

Step 7: If all the tags to be identified have not been successfully identified, jump to start;

Step 8: If there is no tag identification, the flow of algorithm ends.

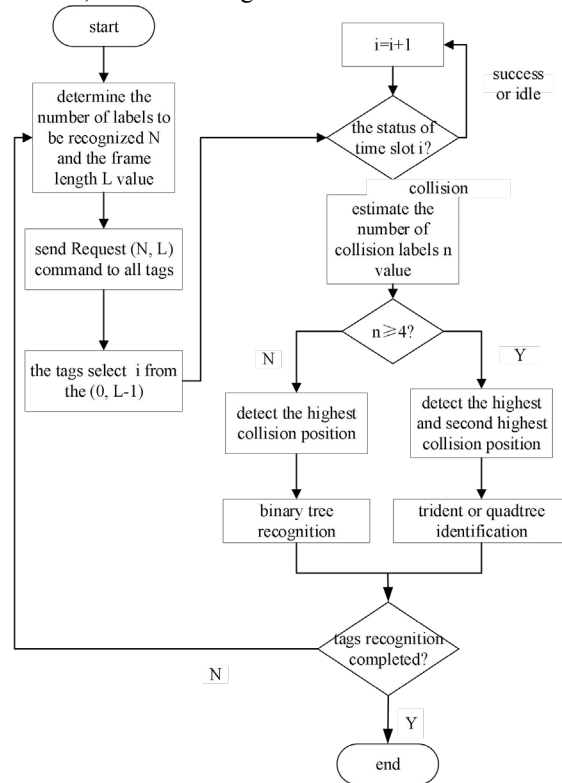


Fig. 3. HMAT algorithm flow chart

#### 2.4. Demo of HMAT algorithm

Suppose there are 10 tags to be recognized within the reader's reading range, and the ID code length is 8 bits, which are TagA: 10000110, TagB: 11010010, TagC: 01000010, TagD: 01001010, TagE: 10010010, TagF: 00001010, TagG: 10110011, TagH: 01010101, TagI: 01101100, TagJ: 00110011.

The recognition process is shown in Figure 4:

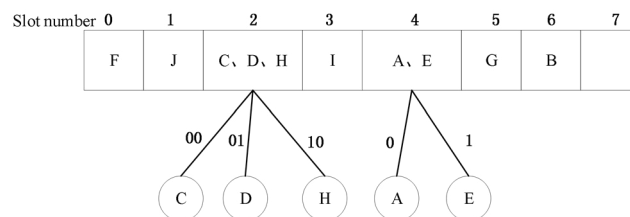


Fig. 4. HMAT algorithm example flow char

The recognition process using HMAT algorithm is as follows:

Step1: Since N equals 10, L equals 8 that can be calculated from formula (1);

Step2: Detect the highest collision bit to get the content after the highest collision bit, then select the response time slot number according to the three bits of information, such as  $(100)_A = (100)_E = 4$ , so tags A and E

respond In time slot 4, it can be calculated that the tag F response is in time slot 0, the tag J response is in time slot 1, the tags C, D and H are in time slot 2, and the tag I is in time slot 3, Tag G responds in time slot 5 and Tag B responds in time slot 6, successfully identifying tags B, F, G, I and J;

Step3: According to the step2, it can be obtained that there are 3 collision tags in time slot 2 which are tags C, D and H, and there are 2 collision tags in time slot 4 which are tags A and E respectively;

Step4: Since there are 3 collision tags in time slot 2, the reader then detects the highest next-highest collision bit, and the collision content is: "00", "01" and "10". The tri-tree algorithm is used to identify and successfully identify the tag C, D and H.

Step5: Since there are two collision tags in time slot 4, which are tags A and E, the card reader detects the highest collision bit, then collision content is: "0" and "1". The tags A and E are identified by BT algorithm;

Step6: All tags are recognized, and the algorithm ends.

It can be drawn from Fig. 4 that N equals 10, and the time slots consumed is 13, with only 1 free time slot and 2 collision time slots. And in the multi-tree stage, there are no collision time slots and idle time slots. Therefore, the HMAT algorithm can greatly reduce collision slots, thereby improving the recognition efficiency.

### 3. Algorithm simulation and analysis

In this paper, programming is performed under MATLAB environment to simulate analysis and comparison of HMAT algorithm. The value range of the number N of tags to be recognized is [10 1000], and the final average value is taken 100 times in the simulation experiment, and four sets of simulation experiments are respectively compared: total slot number comparison, collision slot number comparison, communication overhead comparison and throughput comparison.

Figure 5 compares time slots of four different algorithms. It can be drawn from Figure 5 that time slots required by the HMAT algorithm is the least. When tags to be recognized is 1000, time slots required by the HMAT algorithm is approximately 1400, while the total time slots required by the ACT algorithm is approximately 2100, time slots required by the AHT algorithm is approximately 1810, and DFSA is about 5500, so the HMAT algorithm is optimized in the time slots, especially when the number of tags is large, the optimization is more obvious.

Figure 6 compares collision slots for 4 different algorithms. Since the HMAT algorithm proposed in this paper will only generate collision time slots in DFSA algorithm process, it can ensure that the collision time slots are completely avoided during the recognition process of the multi-tree algorithm. It can also be shown in Figure 6 time slots is significantly less than other three algorithms.

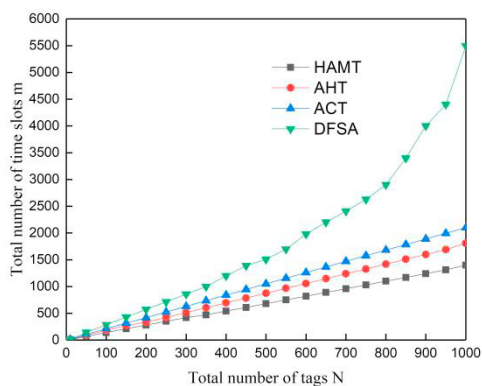


Fig. 5. Time slots of different algorithm

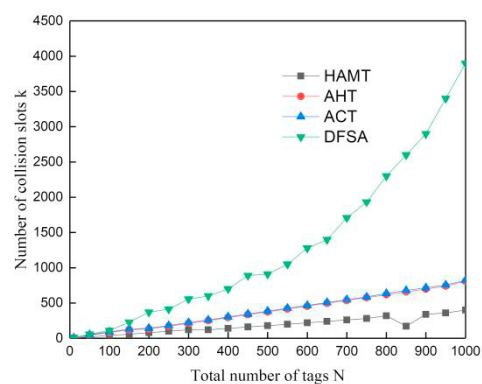


Fig. 6. Collision time slots of different algorithm

Figure 7 is a comparison of recognition efficiency of four algorithms. The recognition efficiency S refers to the ratio of the total number of tags to be identified N to the time slots m required in tag recognition process.

$$S = N/m \quad (2)$$

From Figure 7, the recognition efficiency of the HAMT can reach at around 0.72, which is better than other three algorithms.

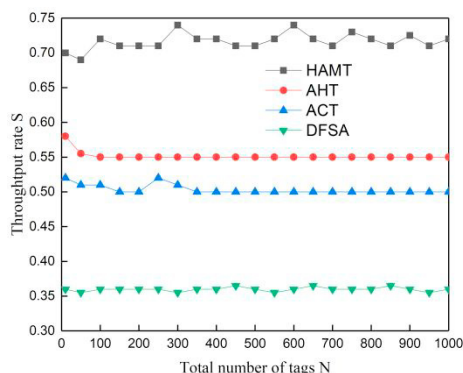


Fig. 7. Throughput of different algorithm.

## 4. Conclusion

Aiming at the shortcomings of binary tree algorithm and ALOHA algorithm, this paper proposes a hybrid algorithm of ALOHA and multi-tree (HAMT). The algorithm combines the advantages of the binary algorithm and the ALOHA algorithm, and is cleverly mixed to achieve 100% tag recognition and speed up the recognition. Simulation results show that HAMT reduces time slots, reduces communication overhead, improves the recognition efficiency, and enables the recognition efficiency to be maintained at about 0.72. Therefore, HAMT can solve the problem of multi-tag collision, and can be promoted and used in the Internet of Things system.

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